

JOSHUA (JOSH) ZHONG

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING

Master of Science in Financial Engineering
GPA: 3.889/4.0

Brooklyn, NY
Expected 05/27

WEST CHESTER UNIVERSITY OF PENNSYLVANIA

Bachelor of Science in Mathematics – Statistics concentration
GPA: 4.0/4.0; Summa Cum Laude; Dean's List (All semesters)

West Chester, PA
08/24

PROGRAMMING / TECHNICAL SKILLS

- **Programming Languages:** Python (QuantLib, Sci-kit learn), C++, SQL, R (Tidyverse, ggplot2), Excel, Java
- **Statistical & Quantitative Methods:** Regression Modeling, Time Series Analysis, Monte Carlo, Backtesting, Hypothesis Testing, Principal Component Analysis (PCA), Clustering

COURSEWORK HIGHLIGHTS

- **Mathematics & Statistics:** Stochastic Calculus, Mathematical Statistics, Topics in Advanced Statistics, Differential Equations, Time Series & Experimental Design
- **Programming:** Machine Learning in Financial Engineering, Financial Computing, C++ Programming for Financial Engineering
- **Finance & Economics:** Options Pricing, Intro to Derivative Securities, Quantitative Methods in Finance

EXPERIENCE

VIEWS DIGITAL MARKETING, Norristown, PA

12/23 – 05/25

Data Analyst, AI Model Development

- Developed and fine-tuned large language models for structured data analysis, implementing prompt engineering to create custom pipelines leading to a 5% increase in conversion rate from leads
- Devised and executed A/B testing frameworks to statistically assess model-driven outcomes and guide iterative refinement

PROJECTS

NEW YORK UNIVERSITY, Brooklyn, NY

Cross-Currency Basis Dynamics in Stablecoin Markets – IAQF Competition Winner

01/26 – 03/26

- Constructed a cross-currency basis framework using minute-level order book snapshots across Kraken, Binance, and Coinbase, documenting a 1,452 bps peak USDC dislocation and a persistent -25.8 bps USDT safe-haven premium bounded by maker/taker fee thresholds; validated cross-venue with pairwise correlation of 0.94
- Modeled arbitrage persistence via AR(1) estimation and ADF unit-root testing, showing USDC basis half-life expanded from 3.7 minutes to 12+ hours during stress consistent with a limits-of-arbitrage breakdown
- Decomposed order book microstructure across three crisis snapshots, revealing one-sided bid depth accumulation in BTC/USDC distinct from balanced BTC/USDT books. Identified Bitcoin's role as an escape vehicle from USDC to USDT

Case Study: Predicting Palladium Futures Prices

09/25 – 12/25

- Built 5 learning models (XGBoost, Random Forest, BiLSTM, LSTM-Attention, ARIMAX) for palladium futures forecasting using 22 engineered features; achieved 45% RMSE improvement over literature benchmarks
- Conducted regime-based robustness analysis across Pre-COVID, COVID-19, and Russia-Ukraine periods; identified model degradation patterns ($2-12\times$ RMSE increase) during structural market breaks

WEST CHESTER UNIVERSITY, West Chester, PA

Modeling California Housing Markets using Multiple Linear Regression

12/23

- Engineered advanced geospatial features by replacing raw coordinates with K-means clusters and distance-to-coast/city metrics, increasing predictive accuracy and interpretability of regression outputs
- Applied Box-Cox transformation and bootstrap resampling to correct heteroscedasticity and produce stable coefficient estimates with reliable confidence intervals

LEADERSHIP / HONORS

- **FRE Department Club (Bulls & Bears), President** 10/25 – Present
- **Conference on AI, Ethics, and Society 2026, Program Committee Member** 05/26 - Present
- **IAQF Academic Affiliate Member Paper Competition, Award Winner** 05/26
- **WCU American Statistical Association, President** 08/23 – 05/24

U.S. Citizen